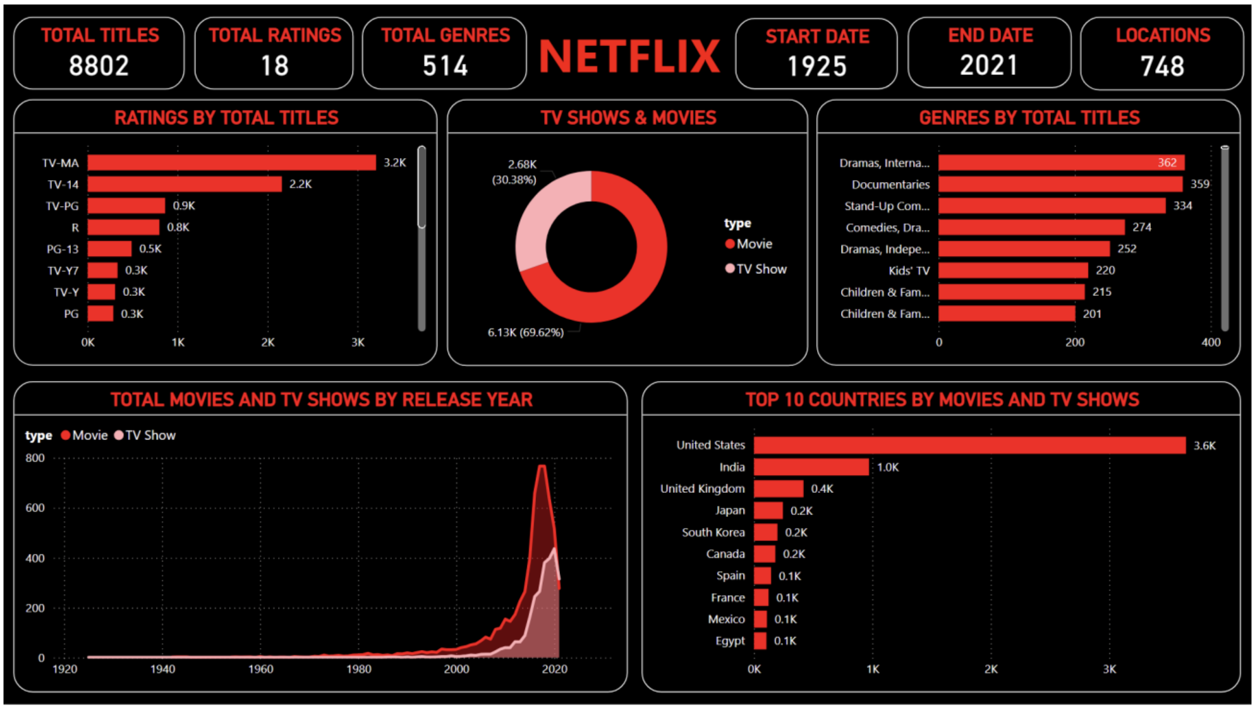
NETFLIX DATA ANALYSIS

Interactive Business Intelligence Dashboard Using Power BI



Project Type	Data Analytics / Business Intelligence
Tool	Microsoft Power BI
Dataset Focus	Netflix Movies and TV Shows
Dashboard Records	8,802 titles

1. Executive Summary

This project presents an interactive Power BI analysis of a Netflix titles dataset. The dashboard converts catalog-level information into a concise set of KPIs and visualizations covering title volume, content type, ratings, genres, release-year trends, and countries.

The dashboard reports 8,802 total titles, 18 distinct ratings, 514 genre combinations, and 748 locations.

The catalog spans release years from 1925 to 2021. Movies represent the larger share of the catalog at approximately 69.62%, while TV Shows account for approximately 30.38%.

The analysis also highlights the dominance of the United States as the leading country by title count, followed by India and the United Kingdom. Recent release years show a substantial increase in title volume, indicating a major expansion of Netflix content in the modern streaming era

Dashboard KPI Snapshot

METRIC	Dashboard Value	Interpretation
Total Titles	8,802	Overall catalog size represented in the dashboard
Total Ratings	18	Distinct rating categories
Total Genres	514	Genre combinations/categories represented
Start Date / Year	1925	Earliest release year shown
End Date / Year	2021	Latest release year shown
Locations	748	Geographic location entries represented

2. Project Objectives

• Analyze the overall composition of the Netflix content catalog.

• Compare Movies and TV Shows by title count.

• Identify the most common rating categories.

• Explore the distribution of titles across genres.

• Study how the number of titles changed across release years.

• Identify the countries contributing the highest number of titles.

• Present the findings through an interactive and business-friendly Power BI dashboard.

Business Questions

• What is the size and composition of the Netflix catalog?

• Are Movies or TV Shows more dominant?

• Which rating categories contain the most titles?

• Which genre combinations are most common?

• How has Netflix content volume changed over time?

• Which countries contribute the largest number of titles?

3. Tools & Technologies

Tool / Technology	Purpose
Microsoft Power BI	Data modeling, DAX-based KPIs, interactive dashboarding and visualization
Power Query	Data preparation and transformation workflow
DAX	Measures and analytical calculations for dashboard KPIs
Netflix Titles Dataset	Source data containing title-level movie and TV-show attributes

4. Analytical Methodology

The project follows a standard business-intelligence workflow: data preparation, modeling, KPI creation, visual analysis, and insight generation

Stage	Description
1. Data Import	Load the Netflix title-level dataset into Power BI.
2. Data Preparation	Review fields, standardize values where required, and prepare the data for analysis.
3. Data Modeling	Organize the analytical fields needed for title, type, rating, genre, year, and location analysis.
4. KPI Development	Create summary measures for title count and distinct categories.
5. Visualization	Build cards, bar charts, donut chart, and time-series visuals.
6. Insight Generation	Interpret content mix, ratings, genres, release-year patterns, and country distribution.

5. Dashboard Design

The dashboard uses a dark background with Netflix-inspired red accents. KPI cards are placed at the top for quick scanning, while the main analytical visuals are arranged beneath them. This structure supports both executive-level overview and deeper exploratory analysis.

• KPI cards provide immediate visibility into catalog size and category counts.

• Horizontal bar charts make rating and country comparisons easy to read.

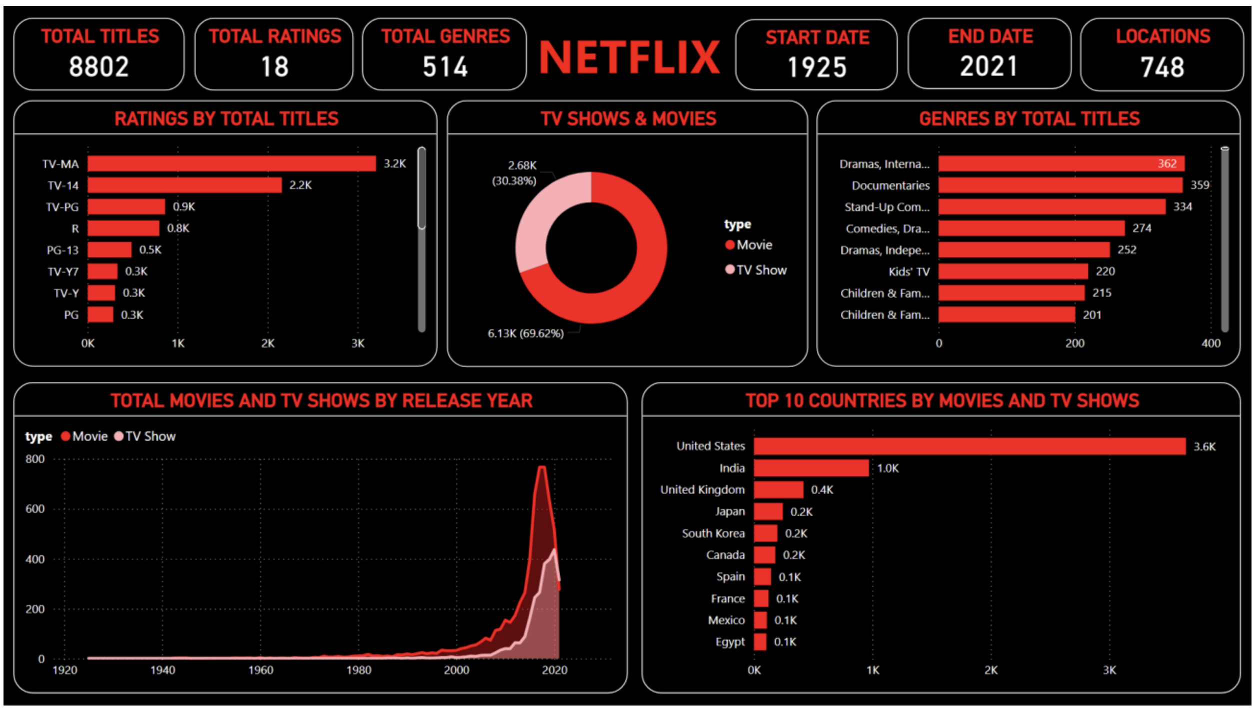
• The donut chart communicates the Movies-versus-TV-Shows content mix.

• The release-year trend visual highlights changes in content volume over time.

• Consistent red-and-black styling provides visual identity and clear hierarchy.

6. Dashboard Overview

The following dashboard is the primary output of the project. It combines six KPI cards/summary indicators with multiple analytical visuals to provide a single-page overview of the Netflix catalog.



7. Key Findings

7.1 Content Type Distribution

Movies form the majority of the catalog, with approximately 6.13K titles (69.62%), while TV Shows account for approximately 2.68K titles (30.38%). This indicates that the analyzed catalog is weighted more heavily toward movie content.

7.2 Ratings Distribution

TV-MA is the largest rating category in the dashboard at about 3.2K titles, followed by TV-14 at about 2.2K and TV-PG at about 0.9K. R-rated titles are also significant at about 0.8K. The distribution suggests that mature and teen-oriented content represents a substantial portion of the catalog.

7.3 Genre Distribution

The leading genre combination shown is 'Dramas, International Movies' with 362 titles, followed closely by Documentaries with 359 titles and Stand-Up Comedy with 334 titles. Several other drama, comedy,children's, and family-oriented combinations also appear prominently.

7.4 Release-Year Trend

The time-series visualization shows relatively low title volume in earlier decades, followed by strong growth from the 2000s onward. The largest concentration occurs in the late 2010s, after which the number of titles shown drops toward 2021. This pattern reflects the strong expansion of streaming-era content.

7.5 Country Distribution

The United States is the leading country by a wide margin at approximately 3.6K titles. India follows at about 1.0K, with the United Kingdom around 0.4K. Japan, South Korea, Canada, Spain, France, Mexico, and Egypt also appear in the top-ten country ranking.

8. Business Recommendations

• Use country-level analysis to evaluate regional content strength and identify markets with growth potential.

• Track Movies and TV Shows separately in future dashboards because their content and audience dynamics differ.

• Monitor genre performance over time rather than relying only on total catalog size.

• Add user engagement, viewing hours, completion rates, or ratings from a separate business dataset to move from catalog analysis toward performance analysis.

• Create interactive filters for year, country, genre, rating, and content type to support deeper exploration.

• Add trend-based KPIs such as year-over-year growth and the share of new releases to strengthen management reporting.

9. Limitations

• The dashboard is primarily catalog-focused; title counts do not directly measure popularity or business performance.

• The available dashboard does not show viewer engagement, revenue, watch time, or user-level behavior.

• Country and genre values can represent multiple associations for a title, so category counts should not always be interpreted as mutually exclusive.

• The release-year analysis ends at 2021 in the displayed dashboard.